



GEV Hazard Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 21-March-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — GEV Hazard Model (MKM-GH-001)

Tests run on **21 March 2026 at 13:30:41**.

Table 1: Test Results Summary — GEV Hazard Model (MKM-GH-001)

Total Tests	16
Passed	16
Failed	0
Skipped	0
Duration	0.27s

Table 2: Detailed Test Results — GEV Hazard Model

Test Description	Acceptance Criteria	Result	Time
Exceedance probability in range	0.0 <= prob <= 1.0	Pass	0.000s
Fit returns three floats	isinstance(shape, float); isinstance(loc, float) (+1 more)	Pass	0.013s
Force gumbel sets shape zero	shape == 0.0	Pass	0.000s
Higher threshold lower probability	p_high < p_low	Pass	0.000s
Return level increases with period	r1_100 > r1_10	Pass	0.000s
Compute all responses returns dict	isinstance(responses, dict); len(responses) == len(synthetic_flat_gauges) (+1 more)	Pass	0.000s
Compute response returns gauge re- sponse	isinstance(response, GaugeResponse); response.gauge_id == gauge_id (+1 more)	Pass	0.000s
Response threshold flags	response.exceeded.alert; response.exceeded.warning	Pass	0.000s
Annual flood probabilities in range	0.0 <= curve.annual_flood_prob.alert <= 1.0; 0.0 <= curve.annual_flood_prob.warning <= 1.0 (+1 more)	Pass	0.084s
Build returns dict of curves	isinstance(curves, dict); len(curves) == len(synthetic_flat_gauges) (+2 more)	Pass	0.081s
Each curve is gauge hazard curve	isinstance(curve, GaugeHazardCurve); curve.gauge_id == gauge_id	Pass	0.090s
Cumulative prob increasing	probs[i] >= probs[i - 1]	Pass	0.000s
Probabilities sum to one	abs(point.survival_prob + point.prob.at.least.one - 1.0) ...	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Returns correct length	<code>len(ts) == 5</code>	Pass	0.000s
Returns term structure points	<code>isinstance(point, TermStructurePoint)</code>	Pass	0.000s
Survival prob decreasing	<code>survivals[i] <= survivals[i - 1]</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (16 tests)	D. Kelly
21-Mar-2026	1.1	16 test(s) added; 16 test(s) removed (total: 32)	D. Kelly